The mahjong-tiles Package
Typesetting Riichi Mahjong Hands and Discard Rivers

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Abstract

mahjong-tiles typesets Japanese/Riichi Mahjong tiles from compact MPSZF notation. It provides a document command, `\mahjong`, for hands and melds, and `\mahjongriver` for discard rivers,and `\stick` for score-stick images. It supports red fives, face-down and blank tiles, rotated and stacked tiles, concealed-kong shorthand, local and global size options, and recolouring of tile backs.

1 Loading the package

A minimal document looks like this.

If you are working from the CTAN source archive, generate the runtime package file first:

```
latex mahjong-tiles.ins
```

The distributed user manual lives in `doc/mahjong-tiles-doc.pdf`;

Its source is `doc/mahjong-tiles-doc.tex`;

The package implementation source is `mahjong-tiles.dtx`;

And `mahjong-tiles.ins` extracts `mahjong-tiles.sty`.

```
\documentclass{article}
\usepackage{mahjong-tiles}
\begin{document}
\mahjong{111m456s111p11122z}
\end{document}
```

The package expects the PDF artwork in the `tiles/` directory. With the default setting, `mahjong-tiles.sty` and `tiles/` should be installed in the same TeX-searchable package directory.

2 Package options

Options can be supplied when loading the package or changed later with `\mahjongtilessetup`.

```
\usepackage[
  height=1.5\baselineskip,
  scale=0.75,
  color=blue!70!black,
  aka=1
]{mahjong-tiles}

\mahjongtilessetup{height=1.5\baselineskip,scale=0.75,color=teal!65!black,aka=1}
```

Option	Default	Description
`height`	`\baselineskip`	Height of one upright tile.
`scale`	`0.75`	Scale factor for the tile symbol relative to the tile face.
`tile-dir`	`tiles`	Directory containing the tile PDF files.
`color`	`none`	Recolour the tile back. Use any `xcolor` colour expression, or `none` for the embedded artwork.
`aka`	`1`	Red-five mode: `0` disables red fives and maps `0m`, `0p`, and `0s` to regular fives; `1` renders them normally; `2` also makes `55555p` show an extra red five pin in concealed-kong shorthand.
`river-cols`	`6`	Number of tiles per discard-river row.
`river-row-gap`	`0pt`	Vertical gap between discard-river rows.
`stick-dir`	`assets/stick`	Directory containing the score-stick PDF files.
`stick-height`	`\baselineskip`	Height of one upright score-stick image.
`stick-scale`	`.5`	Scale factor applied to score-stick images.
`stick-sep`	`.5em`	Horizontal space between different score-stick groups.

3 The `\mahjong` command

```
\mahjong{<tiles>}
\mahjong[<height>][<scale>][<back-colour>][<local-keys>]{<tiles>}
\mahjong[<local-keys>]{<tiles>}
```

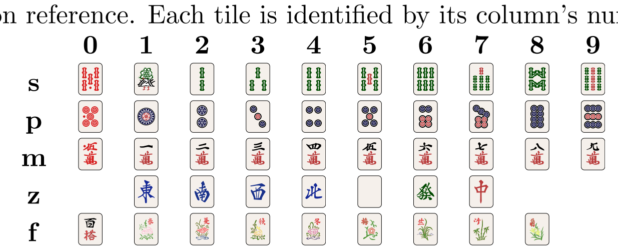
The first optional argument also accepts a key-value list. This allows compact local overrides such as:

```
\mahjong[height=1.5\baselineskip,scale=0.75,color=teal!65!black,aka=0]{x 0m0p0s}
```

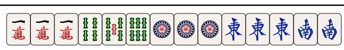
4 MPSZF notation

A tile is written as one or more digits followed by a suit letter. The suits are `m` for manzu, `p` for pinzu, `s` for souzu, `z` for honors, and `f` for flower tiles.

Table 2: MPSZF notation reference. Each tile is identified by its column's number and its row's letter.



The table below shows some examples of combinations.

Input	Output
`111m456s111p11122z`	
`1112345678999p`	
`19m19s19p1234567z`	
`012345678f`	

`123m55p12f`	

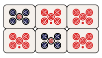
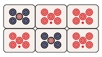
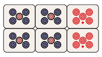
4.1 Special tokens

Token	Meaning
`0m`, `0p`, `0s`	Red fives: 
`x`	Face-down tile: 
`?`	Unknown blank tile: 
`-`	Full visual gap between groups.
`N-`	Proportional gap of `N/7` of one tile width, for example `2-`.

4.2 Rotated and stacked tiles

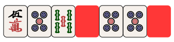
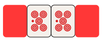
The marker `*` or an apostrophe rotates the preceding tile sideways. For ordinary tiles, the marker `+` or a double quote stacks two sideways copies of the preceding tile, which is useful for kan notation.

There is one red-five edge syntax for upgraded-kan notation. With `aka` enabled, `0m"`, `0p"`, and `0s"` produce a mixed sideways stack with the red five at the bottom and the regular five at the top. Repeating the stack marker twice, as in `0p""` or `0p++`, flips the red five to the top. Three markers, `0p"""` or `0p+++`, force two red fives and deliberately ignore `aka`. The same rule applies to `m`, `p`, and `s`.

Input	Output
`111m111s111p22z2-3*333z`	
`123m-xx?4p'5p"0s`	
`0p" 0p"" 0p"""`	
`0p+ 0p++ 0p+++`	
`aka=0: 0p" 0p"" 0p"""`	

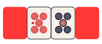
4.3 Red fives and the `aka` option

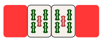
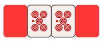
Red fives are controlled by the integer `aka` option. The default `aka=1` renders `0m`, `0p`, and `0s` as red fives. Set `aka=0` to suppress red fives and render those inputs as regular fives. Set `aka=2` to keep normal red fives and add one more red five pin in the `55555p` concealed-kong shorthand.

Input	Output
`0m0p0s55555m`	
`\mahjong [aka=0]{0m0p0s55555p}`	
`\mahjong [aka=2]{55555p}`	

4.4 Concealed kongs

Five identical consecutive digits followed by a suit are rendered as a concealed kan: a face-down tile, two visible tiles, and a face-down tile. For suited fives, the visible pair is controlled by `aka`: `aka=0` gives two regular fives, `aka=1` gives a red five plus a regular five, and `aka=2` gives `55555p` two red five-pin tiles.

Input	Output
`55555p`	

33333z	
\mahjong [aka=0]{55555s}	
\mahjong [aka=2]{55555p}	

For compatibility with older documents, the deprecated option `no-aka` is still accepted. It is equivalent to `aka=0` and emits a package warning. New documents should use `aka=0` directly.

```
\mahjong[aka=0]{0m0p0s}
% Deprecated:
\mahjong[no-aka=1]{0m0p0s}
```

5 Tile-back recolouring

The tile-back colour can be set globally or locally. The colour value is passed to xcolor. The special value `none` restores the embedded back tile.

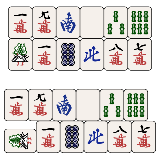
```
\mahjongtilessetup{color=purple!70!black}
\mahjong{x x x}

\mahjong[color=teal!65!black]{x x x}
\mahjong[color=none]{x x x}
```

6 Discard rivers

The `\mahjongriver` command uses the same notation as `\mahjong`, but it breaks tiles into rows. The row length is controlled by `river-cols`.

```
\mahjongriver{1m9m2z5z3s6s1s1m8p4z87m}
\mahjongriver[river-cols=6]{1m9m2z5z3s6s1s*1m8p4z87m}
```

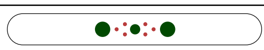
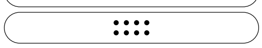


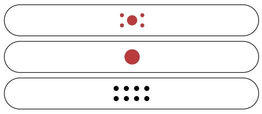
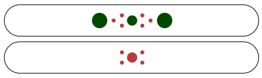
7 Score sticks

The `\stick` command typesets Japanese Mahjong score sticks. The argument may be a three-digit shorthand amount, or a manual specification of stick counts.

A three-digit input is interpreted as hundreds of points. For example, 305 represents 30500 points. The package decomposes the value into an optimal combination of 10k, 5k, 1k, and 100 sticks.

The default artwork directory is `assets/stick/`. It should contain `10k.pdf`, `5k.pdf`, `1k.pdf`, and `100.pdf`.

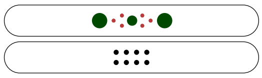
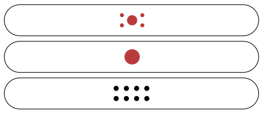
Input	Output
\stick {305}	 × 3  × 5

`\stick {089}`	 $\times$ 1 $\times$ 3 $\times$ 9
`\stick {250}`	 $\times$ 2 $\times$ 1

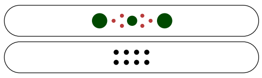
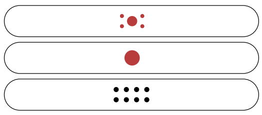
For `305`, the automatic decomposition is:

```
10k \ctimes 3
100 \ctimes 5
```

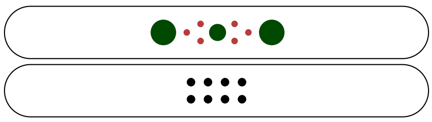
Manual input is also supported. The key-value form names each stick type explicitly.

Input	Output
`\stick {10k=3,100=5}`	 $\times$ 3 $\times$ 5
`\stick {5k=1,1k=2,100=4}`	 $\times$ 1 $\times$ 2 $\times$ 4

The compact comma-list form gives the counts in the order `10k`, `5k`, `1k`, `100`.

Input	Output
`\stick {3,0,0,5}`	 $\times$ 3 $\times$ 5
`\stick [manual]{0,1,2,4}`	 $\times$ 1 $\times$ 2 $\times$ 4

The command accepts local options. These options affect only the current `\stick` call.

Input	Output
`\stick [height=2em,scale=1]{305}`	 $\times$ 3 $\times$ 5

The score-stick defaults can also be changed globally:

```
\mahjongtilessetup{
  stick-dir=assets/stick,
  stick-height=1.2em,
  stick-scale=1,
  stick-sep=.6em
}
```

If the artwork is installed somewhere else, set `stick-dir` accordingly:

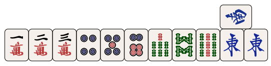
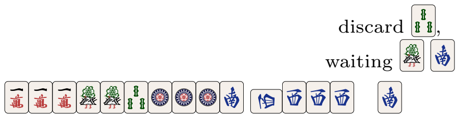
```
\mahjongtilessetup{stick-dir=/assets/stick}
```

8 Overlay notes

The `\mahjong` command supports text annotations. Use `[]` in `\mahjong` to attach a note above the previous tile.

Annotations may contain ordinary TEX material. Mahjong tiles inside an annotation are rendered only when they are explicitly wrapped in `\mj`. For example, `[waiting \mj{3m}]` prints the text "waiting" followed by a small mahjong tile in the annotation. The `\mj` command is parsed only one level deep inside annotations.

The `\mj` command may also be used directly inside `\mahjong`. In that case, its argument must contain exactly one tile. The tile is typeset as a normal-size rotated tile above the hand, which is useful for showing a drawn tile.

Input	Output
`111m111s111p2z2-3*333z7-2z[waiting]`	
`123m456p789s\mj {2z}11z`	
`111m113s111p2z2-3*333z7-2z[discard \mj {3s},\\ waiting \mj {1s2-2z}]`	
`33333z[concealed kong]`	

When a note is added to the fifth tile of a concealed kong shorthand, a warning `Overlay text '<overlay text>' is attached to the final back tile of a concealed kong.` will be triggered. Since a concealed kong is rendered as four tiles, the note is moved to the fourth rendered tile, namely the final back tile.

9 Expl3 interface

For package authors who prefer an expl3-style interface, mahjong-tiles also exposes:

```
\ExplSyntaxOn
\mahjongtiles_typeset_hand:n {111m456s111p11122z}
\mahjongtilesriver:n         {111m456s111p11122z}
\ExplSyntaxOff
```

10 Attribution and licence

The LaTeX package code is distributed under the MIT Licence. Portions are based on Daniel Schmitz's **mahjong** package, copyright 2021 Daniel Schmitz, also under the MIT Licence.

The tile artwork in `tiles/` is derived from FluffyStuff's `riichi-mahjong-tiles` project. The upstream licence file places that work in the public domain/CC0. Attribution is retained here and in the README.